
Yeshan Qiu

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EDUCATION

National University of Singapore (NUS) Singapore, Singapore 01/2022-07/2026

PhD in Architecture, degree requirements completed; conferral pending
Thesis successfully defended on 30 July 2026 | Research focus: Landscape ecology
Thesis: Understanding the Impact of Urban Patterns on Urban Avian Biodiversity: Comparative Studies Across Cities

Shanghai Jiao Tong University (SJTU) Shanghai, China 09/2018-03/2021

Master's Degree in Landscape Architecture
Dissertation: Construction and Design Application of Bayesian Model of Public Ecological Aesthetic Perception in Urban Green Space Based on Deep Learning

Hainan University (HNU) Haikou, China 09/2014-06/2018

Bachelor's Degree in Landscape Architecture
Overall GPA: 3.82/4.0 Ranking: 1/136

RESEARCH INTERESTS

- Urban biodiversity, avian ecology, and biodiversity modelling
- Urban landscape patterns, 3D vegetation structure, and pattern–process relationships
- Ecological aesthetics, human perception, and ecosystem service assessment

PUBLICATIONS

Manuscripts under review / in preparation for submission

Qiu, Y., Chi, D., Hahs, A. K., Richards, D. R., Yang, J., Meili, N., Zhu, Y., Wang, J., Lin, B. B., Manoli, G., Zhang, Y., Burlando, P., McDonnell, M. J., & Tan, P. Y. (2026). Systematic Review of the Relationships between Multidimensional Urban Patterns and Avian Biodiversity: A Meta-analysis [Manuscript under review].

Qiu, Y., Chi, D., Hahs, A. K., Richards, D. R., Yang, J., Meili, N., Zhu, Y., Wang, J., Lin, B. B., Manoli, G., Zhang, Y., Burlando, P., McDonnell, M. J., & Tan, P. Y. (2026). Mapping Methods Matter: Influence of Spatial Analytical Units on Urban Avian Biodiversity Assessment: A Meta-analysis [Manuscript under review].

Qiu, Y., Chi, D., Hahs, A. K., Richards, D. R., Yang, J., Meili, N., Zhu, Y., Wang, J., Lin, B. B., Manoli, G., Zhang, Y., Burlando, P., McDonnell, M. J., & Tan, P. Y. (2026). Assessing Urban Pattern Effects on Bird Species Richness Across 68 Cities [Manuscript in preparation for submission].

Peer-reviewed journal articles

Qiu, Y., Chi, D., Meili, N., Zhu, Y., Wang, J., McDonnell, M., & Tan, P. Y. (2024). Exploring the Impact of Urban Pattern on Avian Occurrence: A Buffer-Based Analysis Using Citizen Science Data. AGU Fall Meeting Abstracts, 153, B43B-153.

Qiu, Y., Pan, H., Kalantari, Z., Giusti, M., & Che, S. (2023). The natural focus: Combining deep learning and eye-tracking to understand public perceptions of urban ecosystem aesthetics. *Ecological Indicators*, 156, 111181.

Qiu, Y., Che, S. (2022). Construction Strategies for Urban Ecological Green Space Based on Ecological Aesthetics. *Landscape Architecture*, 29(7), 37-43.

Qiu, Y., Xie, C., Pan, H., & Che, S. (2021). Study of Urban Street-Scene and Landscape Aesthetic Public Perception Based on Deep Learning: The Case of Shanghai. *Chinese Landscape Architecture*, 37(6), 77-81.

Qiu, Y., Chen, Y., & Che, S. (2020). Using Eye-tracking and Deep Learning Approach to Promote Naturalness in Urban Landscape. *EGU General Assembly Conference Abstracts*, EGU2020-6515.

Chi, D., Manoli, G., Lin, B., Aerts, R., Yang, J., Hahs, A., Richards, D., Meili, N., **Qiu, Y.,** et al. (2025). Residential tree canopy configuration and mortality in 6 million Swiss adults: a longitudinal study. *The Lancet Planetary Health*, 9(3), e186-e195.

Meili, N., Zheng, X., Takane, Y., Nakajima, K., Yamaguchi, K., Chi, D., Zhu, Y., **Qiu, Y.,** et al. (2025). Modeling the effect of trees on energy demand for indoor cooling and dehumidification across cities and climates. *Journal of Advances in Modeling Earth Systems*, 17(3), e2024MS004590.

Zhu, Y., Wang, J., Manoli, G., Zhang, Y., Chi, D., Meili, N., **Qiu, Y.,** Lin, G. S., Tan, P. Y., Burlando, P., & Fatichi, S. (2025). Influence of urban form and function on daytime-nighttime population differences and hazard risk assessments. *npj Urban Sustainability*, 5, 91. <https://doi.org/10.1038/s42949-025-00282-0>

Chi, D., Meili, N., Zhu, Y., Wang, J., **Qiu, Y.,** Li, C., Burlando, P., Fatichi, S., Lin, B., et al. (2024). Associations between greenspace patterns and mortality in six million Swiss adults: a longitudinal study. *ISEE Conference Abstracts*, 1.

Meili, N., Zheng, X., Chi, D., Zhu, Y., **Qiu, Y.,** Wang, J., Manoli, G., Burlando, P., et al. (2023). Quantifying Vegetation Effects on Climate and Building Energy Demand in Cities. *AGU Fall Meeting Abstracts*, GC44F-01.

Sun, W., **Qiu, Y.,** Yu, B., & Wang, Y. (2024). Inclusive Design of Smart Landscape in Community Parks for The Migrant Elderly. *Human Factors in Design, Engineering, and Computing*, 159(159).

Chen, Y., **Qiu, Y.,** & Che, S. (2020). Combining deep learning and multi-source data to promote subjective perception of ecosystem services in urban landscape. *EGU General Assembly Conference Abstracts*, EGU2020-6516.

Qiu, Y., Xu, X. (2020). Landscape Design for Modern Healing Spaces under the Influence of Traditional Chinese Health Culture. *Chinese Horticulture Abstracts*, 33(7), 106-108

CONFERENCE PRESENTATIONS

Qiu, Y., Chi, D., Zhu, Y., Wang, J., McDonnell, M. J., & Tan, P. Y. (2025). Thinking Vertically: How 3D Urban Form Shapes Global Avian Biodiversity: Multi-City evidence for functional guild-specific responses to 3D urban form. *British Ecological Society Annual Meeting 2025*. Poster presentation.

Qiu, Y., Chi, D., Meili, N., Zhu, Y., Wang, J., McDonnell, M. J., & Tan, P. Y. (2024). Exploring the Impact of Urban Pattern on Avian Occurrence: A Buffer-Based Analysis Using Citizen Science Data. *AGU Fall Meeting Abstracts*, 153, B43B-153. Poster presentation / Conference abstract.

Qiu, Y., Chi, D., Meili, N., Zhu, Y., Wang, J., McDonnell, M. J., & Tan, P. Y. (2024). Exploring the Impact of Urban Pattern Metrics on Avian Occurrence: A Buffer-Based Analysis Using Citizen Science Data. Ecological Society of America Annual Meeting2024. Poster presentation.

RESEARCH EXPERIENCE

Comparative Ecology of Cities: Understanding Patterns, Processes and Properties of Cities to Improve Urban Conditions

03/2023-Present

Future Cities Lab Global, NUS, ETH Zurich

Researcher, Biodiversity Work Package Lead

Project Lead: Prof. Dr. Tan Puay Yok, Prof. Dr. Paolo Burlando

- Contributed to a collaborative, multi-module project examining urban pattern–process–function relationships across cities, linking settlement patterns with biodiversity, climate and energy, hydrology, public health, and socio-economic outcomes.
- Independently led the biodiversity module through a cross-city comparative study of 70+ global cities, integrating citizen-science species occurrence records, biodiversity observation data, and spatial pattern metrics into modelling-ready datasets to assess relationships between urban spatial patterns and biodiversity responses.
- Conducted multi-source, multi-city urban pattern metrics analysis, integrating land-use/land-cover data, urban spatial layers, landscape metrics, and 3D urban/vegetation structure indicators.
- Analysed how landscape distribution, composition, configuration, and three-dimensional structure shape biodiversity responses, with implications for planning practices, ecological networks, and nature-based solutions.

Collaborative Urban Bird Modelling Projects with NParks

07/2024- Present

Wildlife Management Research, National Parks Board, Singapore

Researcher

Collaborator: Dr. Malcolm Soh, Principal Researcher (Wildlife Management Research)

- Collaborated with NParks' Wildlife Management Research team on applied avian biodiversity modelling projects, including the Decadal Study of Birds in Singapore and straw-headed bulbul population modelling.
- Integrated bird survey data with Landsat-derived land-cover layers, LiDAR-derived vegetation structure metrics, ETH green canopy data, and multi-scale landscape metrics.
- Generated modelling-ready spatial predictors from transect buffers, survey points, and prediction grids, including landscape composition, configuration, canopy connectivity, patch structure, total edge, core area, and percentage cover metrics.
- Supported long-term urban bird community analysis and straw-headed bulbul abundance/population modelling for management-oriented biodiversity assessment and urban wildlife planning.

Public Risk Perception, Behaviour Patterns and Public Emotional Guidance under Major Public Health Emergencies

03/2020-2023

National Natural Science Youth Fund Project

Project No.: 7204100421

Researcher

Advisor: Associate Professor Bingqin Yu

- Analysed the impact and potential role of urban green space on public health events and public health;

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- Concluded that the group pressure and psychological restoration of the elderly in the community are strongly correlated with the rate of green space and green vision.

Synergistic Research on the Application of Street Species in Shanghai and the Compound Benefits of High-Quality Road Greening 11/2019-09/2020

Shanghai Science and Technology Commission

Project No.: 19DZ1203702

Team Leader

Advisor: Professor Shengquan Che

- Responsible for the evaluation of benefits of urban street greening in Shanghai using AI deep learning skills;
- Assisted in optimization of ecological design, including increasing the plant landscape (after evaluation) by 23.2%, seasonal colour richness of plant landscape by 33.67%, and overall green space utilization by 8%;
- Led 6 team members to implement preliminary investigations and assessments of ecologic road environments;
- Designed a natural plant community pattern in urban road green space across the Jing'an district of Shanghai. Independently produced a research report;
- Innovatively utilised ecological natural plant community's mode to replace traditional geometric and regular plant communities in road green spaces to improve the ecological stability of green spaces and increase biodiversity.

Survey of Shanghai Urban Landscape and Style: Based on Artificial Intelligence and Big Data 03/2019-11/2020

National Natural Science Youth Fund Project

Project No.: 52000130

Researcher Assistant

Advisor: Dr Haozhi Pan

- Interviewed 86 households face-to-face to investigate public perception of urban landscape aesthetics. Used deep learning, streets data and eye tracking technology;
- Analysed street view data with Python to explore 238 road landscapes in downtown Shanghai;
- Set up a street view sample database to conduct eye-movement experiments on 57 local residents in Shanghai, and tracked their saccades, gazes, glances and other physiological index individually;
- Built aesthetic cognition Bayesian model using deep learning to expand the limited aesthetic survey samples with the aim of constructing a rapid and large-scale general survey of public aesthetic preferences;
- Published the survey to core journal papers *Chinese Landscape Architecture* of Beijing University.

PROFESSIONAL EXPERIENCE

Dawson Landscape Design Co., Ltd.

01/2018-08/2018

Landscape Designer Assistant, Landscape design

- Participated in preliminary analysis of status quo in beautiful village revitalization construction planning and design projects of Haikou city;
- Independently designed Ding'an Community Park planning projects within two weeks and implemented the first draft and final version of the floor plan, which was successfully adopted by clients;

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- Collected data on the site's human history, topography, precipitation, sunlight, and animal and plant biodiversity in the design project of Magnolia Tianxiang Farm. Produced relevant floor plans;
 - Performed modelling analysis by combining GIS and Rhino;
 - Simulated the distribution of light, wind direction and wind speed in four seasons before selecting the best habitat for plants and crops with different living habits;
 - Took full responsibility for the design of the municipal square landscape of Ding'an City, such as site investigation and analysis, floor plan design and design project plan text writing;

SELECTED HONOURS & AWARDS

- **First prize** in Academic Scholarship, SJTU 10/2020
- **Second prize** in Academic Scholarship, SJTU 10/2019
- **Third prize** in the National University Landscape Garden Design Competition, CHSLA 10/2019
- Scholarship of National Academic (awarded to **Top 1%**) 11/2017
- **Third prize** in the National University Student Environmental Ecological Science and Technology Innovation Competition (awarded to **Top 3%**) 09/2017
- Outstanding Graduate Student, SJTU 03/2021
- Outstanding Graduate Student, HNU 2015-2017

ACADEMIC SKILLS

- **Ecological and biodiversity modelling:** biodiversity modelling, ecological response modelling, avian occurrence analysis, and ecosystem service assessment.
- **Geospatial analysis:** GIS, raster/vector processing, land-use/land-cover analysis, landscape metrics, LiDAR-derived vegetation metrics, and multi-source geospatial data integration.
- **Programming and statistical tools:** R, Python, MATLAB, Netica; statistical modelling, Bayesian modelling, reproducible workflows, and data visualisation.
- **Design and experimental tools:** Adobe Photoshop, Adobe Illustrator, SketchUp, Lumion, WUFI, Tobii eye-tracking equipment, and physiological-response data collection.

EXTRACURRICULAR ACTIVITIES

- *Teaching Assistant* for Academic Writing, Integrity and Ethics and Landscape Garden Graduation Design 09/2018-08/2020
- *Leader* of the Bulgarian Rural Research and Planning Practice Project of SJTU's Global Challenge Program 05/2019-08/2019